

Topic-Guided Humor Generation and Evaluation Using GPT-4o-mini

Raghd Hassan Lujain Alhusan Raseel almanea Maha Alnassar Alya Althobiani

King Saud University

College of Computer and Information Sciences

Department of Information Technology

{443204743, 444200785, 444201169, 444200604, 444200973}@student.ksu.edu.sa

Supervised by: HEND ALRASHEED

Abstract

This paper presents a topic-guided humor generation pipeline using GPT-4o-mini. The study uses the Short Jokes Dataset, applies text pre-processing, ethical content filtering, data reduction, topic discovery, and topic mapping before generating one-line jokes across five selected content-based topics: Relationships, Health, Food, Animals, and Politics. The generation stage uses few-shot topic-conditioned prompting, where two topic-related examples are provided before asking the model to generate a new joke. The generated jokes are evaluated mainly through human judgment using three criteria: funniness, topic relevance, and fluency. Automatic humor detection models are used only as supporting analysis to compare their scores with human funniness ratings. The results show high topic relevance and fluency, moderate funniness, and weak alignment between automatic humor scores and human judgment. These findings suggest that GPT-4o-mini can follow topic conditions and produce fluent humorous text, but human evaluation remains necessary for judging humor quality.

1 Introduction

Humor is an important yet challenging aspect of human language because it depends on linguistic creativity, ambiguity, social context, and subjective interpretation (Binsted et al., 2006; Amin and Burghardt, 2020). In daily communication, humor is used to attract attention, reduce tension, express social meaning, and make interaction more engaging. For Natural Language Processing (NLP), humor generation is especially difficult because it requires more than producing fluent and grammatically correct text. A sentence may be clear and well-formed, yet still fail to be funny. Successful humor often depends on surprise, word choice, cultural background, and the relationship between a setup and a punchline. Previous studies in computational humor have also shown that humor is

difficult to recognize and evaluate automatically because funniness is subjective and can vary across people, topics, and contexts (Mihalcea and Strapparava, 2005a; Meaney et al., 2021a).

Recent advances in neural language generation have improved the ability of models to produce coherent and contextually relevant text. Transformer-based architectures have become widely used in NLP because they model relationships between words and contexts through attention mechanisms (Vaswani et al., 2017). Large language models have further improved open-ended generation and instruction-following abilities, making them useful for tasks that require flexible language production (Brown et al., 2020; OpenAI, 2023). At the same time, controllable text generation has become an important research direction because users often need generated text to follow specific constraints such as topic, tone, sentiment, or style (Prabhumoye et al., 2020a; Zhang et al., 2023). Humor generation can therefore be viewed as a special case of controllable generation, where the output must be both humorous and relevant to a given topic.

However, most prior work in computational humor focuses on humor detection, joke classification, sarcasm recognition, or unrestricted humor generation (Mihalcea and Strapparava, 2005a; Amin and Burghardt, 2020). While these studies are valuable, less attention has been given to controlled humor generation across content-based topics, where the same model is evaluated under different social or topical conditions using human judgment. This gap is important because a model may generate fluent text, but still fail to produce humor that is funny, safe, and relevant to the requested context. In addition, automatic humor evaluation remains limited: humor classifiers may identify whether a text looks like a joke, but they may not fully match human judgments of funniness. Prior work on text generation evaluation also highlights that human

evaluation is often necessary because automatic metrics may fail to reflect qualities such as creativity, coherence, and user perception (Celikyilmaz et al., 2020; van der Lee et al., 2021).

In this work, we propose a controlled topic-guided humor generation framework using GPT-4o-mini (OpenAI, 2024). Rather than asking the model to generate random jokes, we condition generation on selected content-based topics, including *Relationships*, *Health*, *Food*, *Animals*, and *Politics*. The goal is not to build a topic classifier for the entire joke dataset. Instead, the goal is to test whether a language model can generate short humorous text that remains fluent, funny, and relevant across different topics and social contexts.

This study is guided by three research questions:

- **RQ1:** Can GPT-4o-mini generate jokes that are generally fluent and humorous?
- **RQ2:** Does the quality of generated humor vary across different topics?
- **RQ3:** Can the model generate jokes that remain relevant to the requested topic or social context?

Our main contribution is the design and implementation of a complete NLP pipeline for topic-guided joke generation and evaluation. First, we prepare a safer working subset from the Short Jokes dataset through cleaning, reduction, and toxicity filtering. Second, we apply topic discovery on an unlabeled subset and map raw topic labels into broader content-based categories. Third, we design a controlled few-shot prompting strategy in which GPT-4o-mini generates new jokes using two topic-specific examples. Finally, we evaluate the generated jokes using five human raters and compare human funniness judgments with automatic humor detection models as a supporting analysis.

The novelty of this work is not in inventing humor generation from scratch, but in applying a controlled topic-guided setup that combines data-driven topic discovery, content-based topic mapping, few-shot generation, and human-centered evaluation. This setup makes it possible to examine whether some topics are easier for the model to generate humor about than others, and whether automatic humor classifiers align with human perception of generated jokes.

The proposed method uses the Short Jokes dataset as the main data source. After cleaning

and ethical filtering, a safe subset is used for topic discovery and topic selection. GPT-4o-mini is then used to generate 50 jokes in total, with 10 jokes for each selected topic. Each generated joke is evaluated by five human raters using three criteria: funniness, topic relevance, and fluency. The results show that the model achieves high relevance and fluency scores across topics, while funniness is more variable. Politics achieves the highest average funniness score, while Food receives the lowest. The comparison between human ratings and automatic humor scores shows weak agreement, suggesting that automatic humor classifiers do not reliably capture human perception of funniness in generated jokes.

The rest of this paper is organized as follows. Section 2 reviews related work on controllable text generation, humor generation, topic discovery, few-shot prompting, and humor evaluation. Section 3 describes the dataset, preprocessing steps, topic discovery process, topic mapping, few-shot generation method, and evaluation design. Section 4 presents the results of the human and automatic evaluations. Section 5 discusses the findings, compares them with prior work, and explains the limitations and social implications of the approach. Finally, Section 6 concludes the paper and suggests possible directions for future work.

2 Related Work

Humor generation sits at the intersection of several active research areas in NLP, including computational humor, controllable text generation, and few-shot prompting with large language models. This section reviews the most closely related work across these three directions.

2.1 Humor Generation and Evaluation

Early work in computational humor focused on rule-based systems that relied on handcrafted templates and linguistic constraints to produce jokes. Mihalcea and Strapparava (2005b) investigated the linguistic properties of humorous text and explored automatic humor recognition, establishing foundational insights into how incongruity, surprise, and ambiguity contribute to the perception of funniness. Their work highlighted that humor is a subjective phenomenon that varies across individuals and social contexts, a challenge that remains central to humor research today.

More recently, systems have moved toward data-

driven and neural approaches. [Toplyn \(2021\)](#) presented Witscript, a joke generation system designed to improvise original, contextually relevant jokes during a conversation. Witscript employs NLP tools to extract keywords from a topic sentence and uses wordplay to link those keywords to a punchline, which is then completed using a language model fine-tuned on television monologue jokes. An internal scoring mechanism filters out low-quality candidates, and human evaluators judged the system’s responses to be jokes more than 40% of the time. Building on this, [Toplyn \(2023\)](#) introduced Witscript 3, a hybrid system that generates joke candidates through three distinct production mechanisms and selects the best output using internal scoring. Human evaluators rated Witscript 3’s responses as jokes in 44% of cases. These systems demonstrate that joke generation in conversational settings is tractable, though producing consistently high-quality humor remains an open challenge.

Beyond generation, evaluating humor automatically has proven equally difficult. [Meaney et al. \(2021b\)](#) organized a shared task on humor detection and rating, revealing that funniness scores vary significantly across annotators and that automatic systems still struggle to match human judgment reliably. [Tikhonov and Shtykovskiy \(2024\)](#) approached the generation problem from a reasoning perspective, proposing a multistep reasoning framework for one-liner joke generation using large language models. Their human evaluation showed that this structured approach consistently improved joke quality compared to direct zero-shot generation, suggesting that decomposing the creative process into explicit reasoning steps is beneficial for humor production. Unlike these prior systems, our work does not aim to generate open-ended jokes but instead focuses on topic-guided generation, where jokes are explicitly constrained to selected content-based topics derived from a real dataset. This allows us to examine whether humor quality varies systematically across topics and whether automatic classifiers agree with human judgments of funniness.

2.2 Topic Discovery with Large Language Models

Traditional topic modeling approaches often represent topics as sets of keywords, which can make the resulting topics difficult to interpret without additional human analysis. Recent work has explored using large language models to support

topic discovery and labeling because LLMs can produce more interpretable natural-language topic descriptions. [Pham et al. \(2024\)](#) introduce TopicGPT, a prompt-based topic modeling framework that uses LLMs to uncover latent topics in text collections and generate meaningful topic labels. This approach motivates our use of GPT-4o-mini for topic discovery, where the model assigns interpretable topic labels to a subset of jokes rather than relying on keyword-based representations that may be harder to map into meaningful generation categories.

This direction is relevant to the present work because the Short Jokes dataset is unlabeled and does not provide topic categories for each joke. Instead of manually choosing topics without evidence from the data, we use GPT-4o-mini to assign short raw topic labels to a discovery subset. These labels are then normalized and mapped into broader content-based categories. The goal is not to build a supervised topic classifier, but to derive clear generation topics from the dataset itself.

2.3 Controllable Text Generation

Controllable text generation has become an important research direction in NLP because real-world applications often require generated text to conform to specific attributes such as topic, sentiment, or style. Early work in this area explored style transfer between non-parallel corpora ([Prabhumoye et al., 2020b](#)), where the goal is to change surface-level attributes of a text while preserving its underlying content. [Liang et al. \(2024\)](#) provided a broader survey of controllable generation methods and their evaluation challenges, noting that controlling fine-grained attributes such as topic and tone simultaneously remains difficult for current models.

Humor generation can be viewed as a special case of controllable text generation, where the model must produce output that satisfies both a topical constraint and a funniness objective. Prior work has shown that conditioning generation on explicit attributes, such as sentiment ([Prabhumoye et al., 2020b](#)) or reframing strategy ([Ziems et al., 2022](#)), can improve the relevance and quality of generated text. [Ziems et al. \(2022\)](#) introduced a positive reframing task in which models are trained to rewrite negative text using theoretically motivated strategies from positive psychology, demonstrating that conditioning generation on strategy labels leads to more meaningful and fluent outputs

than unconstrained generation. This line of work motivates our use of topic labels as explicit constraints during joke generation. In this work, we adopt a controlled few-shot prompting strategy that fixes the topic for each generation condition, allowing direct comparison of humor quality across different topic categories.

2.4 Few-Shot Prompting with Large Language Models

The introduction of GPT-3 by [Brown et al. \(2020\)](#) demonstrated that large language models can perform new tasks from only a small number of input-output demonstrations, without any gradient updates or fine-tuning. This few-shot learning paradigm has since been widely adopted across NLP tasks that require flexible and creative language production. In text generation tasks specifically, providing in-context examples has been shown to guide models toward the desired output format and style more effectively than zero-shot prompting alone.

In humor generation, few-shot prompting has been used to supply the model with examples that establish the expected joke style and structure before requesting new output ([Tikhonov and Shtykovskiy, 2024](#)). Similarly, [Ziems et al. \(2022\)](#) evaluated few-shot GPT-3 and GPT-Neo on a text reframing task and found that, while few-shot systems achieved comparable sentiment shift to supervised models, they were less effective at preserving the meaning of the original text. This finding suggests that few-shot prompting alone may not be sufficient for tasks that require both creativity and constraint adherence. Our work applies a similar few-shot strategy, providing two topic-specific joke examples in each prompt to guide GPT-4o-mini toward generating jokes that are both humorous and relevant to the requested topic ([OpenAI, 2024](#)). The prompt structure is kept identical across all topic conditions to ensure a fair comparison, and the impact of topic on generation quality is evaluated through both human ratings and automatic humor classifiers.

3 Method

3.1 Dataset

We used the Short Jokes Dataset, a large English humor dataset containing 231,657 short jokes in CSV format ([Moudgil, 2017a](#)). Each record has two fields: ID, which represents the unique joke

Item	Value
Original records	231,657
Columns	ID, Joke
Language	English
After preprocessing	227,193
Reduced sample	2,000
After toxicity filtering	1,702
Discovery subset	300
Generated jokes	50

Table 1: Summary statistics for the dataset and experimental subsets.

identifier, and Joke, which contains the joke text. Since the dataset contains short joke texts, it is suitable for this project, which focuses on generating one-line jokes rather than long narrative humor.

The dataset was collected from multiple online sources containing short jokes, including joke websites, humorous tweet sources, Reddit joke posts, and an existing SQL joke file ([Moudgil, 2017b](#)). The repository reports that Python scraper scripts were used to extract jokes from different sources and store them in a unified CSV format. For regular web pages, the scripts used web scraping tools, while the Reddit portion used Reddit-related extraction. The clearly reported collection period applies only to the Reddit portion, which covers posts from 25 January 2008 to 31 January 2017. For the remaining web sources, exact collection dates are not explicitly reported; therefore, we do not assume that the full dataset covers the same period.

Table 1 summarizes the main dataset statistics used in this project.

3.2 Preprocessing and Ethical Filtering

Because the dataset was collected from web sources, it may contain inappropriate or offensive content. The dataset card also warns that some offensive jokes may remain despite attempts to keep the dataset clean ([Moudgil, 2017a](#)). Therefore, we did not use the raw dataset directly. We first applied preprocessing using Pandas and regular expressions. The cleaning steps removed missing values, duplicate jokes, empty strings, URLs, corrupted encoding symbols, symbol-only rows, excessive repeated characters, and jokes with a high symbol ratio. We also applied a word-count filter and kept jokes between 6 and 40 words. This range was chosen because extremely short texts may lack enough context, while very long texts are

less suitable for one-line joke generation. After pre-processing, the dataset was reduced from 231,657 to 227,193 jokes.

After cleaning, we randomly sampled 2,000 jokes using a fixed seed (`random_state=42`) to make the experiment reproducible and computationally manageable. Since technical cleaning does not guarantee that the text is safe, we applied toxicity filtering using the Detoxify original model (Unitary AI, 2020). Detoxify predicts multiple harmful-content dimensions, including toxicity, severe toxicity, obscene language, insult, identity attack, and threat. This is important for joke data because humorous text may contain sarcasm, harsh wording, or sensitive references.

We used a threshold of 0.6 for all toxicity categories. A joke was kept only if all toxicity-related scores were below 0.6. This threshold was selected as a practical balance between reducing inappropriate content and preserving joke diversity. The data was processed in batches of 128 for efficiency. After toxicity filtering, the final safe subset contained 1,702 jokes, which was used for topic discovery and later generation-related steps.

3.3 Topic Discovery

For topic discovery, we sampled 300 jokes from the safe subset using a fixed seed. This subset was used to identify common themes in the cleaned and filtered data, not to label the entire corpus. The smaller subset made the topic analysis computationally manageable while still allowing repeated patterns to appear.

We used GPT-4o-mini to assign one short topic label to each joke. This choice was motivated by recent work showing that large language models can support interpretable topic labeling by producing natural-language labels rather than only keyword clusters (Pham et al., 2024). The 300 jokes were processed in batches of 20, resulting in 15 batches. For each batch, the model was instructed to identify the main topic of each joke and return one short label without explanation. We used temperature = 0 because topic discovery is closer to annotation than creative generation. This step produced 300 discovered labels, including labels such as *wordplay*, *puns*, *politics*, *food*, *relationships*, *dark humor*, *dating*, *health*, and *animals*.

3.4 Topic Mapping and Selection

After topic discovery, the raw labels were normalized before final topic selection. This step was

Final Topic	Frequency
Relationships	28
Health	24
Food	17
Animals	17
Politics	16

Table 2: Final selected topics and their frequencies after topic mapping.

necessary because the model produced labels with formatting variation, such as *Wordplay* and *word-play*. We normalized the labels by converting them to lowercase, removing extra spaces, and replacing hyphens with spaces. Then, fine-grained labels were manually mapped into broader content-based categories. For example, *flirtation*, *dating*, *marriage*, and *romance* were mapped to *Relationships*, while *psychology*, *mental health*, *illness*, and *disease* were mapped to *Health*.

Labels that described joke style rather than joke content, such as *wordplay*, *puns*, *humor*, and *absurdity*, were mapped to *General*. This decision was made because these labels describe how the joke works, not what the joke is about. Since this project uses content-based topics for generation, *General* was excluded from final topic selection. The *Adult* category was also excluded because the project aims to generate clean, general-audience jokes.

After excluding *General* and *Adult*, the most frequent usable categories included Relationships, Health, Food, Animals, Crime, Politics, Entertainment, and Family. Although Crime had the same frequency as Politics, Politics was selected instead because it better matched the project goal of comparing social and content-based contexts while avoiding a topic that may encourage violent or unsafe humor. The final selected topics were Relationships, Health, Food, Animals, and Politics. Table 2 shows the selected topics and their frequencies in the discovery subset.

3.5 Few-Shot Joke Generation

After selecting the five topics, we used GPT-4o-mini to generate new topic-guided jokes. GPT-4o-mini was selected because the project required repeated generation calls, instruction following, and short text generation (OpenAI, 2024). The model was not fine-tuned; all generation was performed through prompting.

We used a few-shot prompting approach. For each topic, two jokes from the same topic were included in the prompt, and the model was asked to generate a new joke related to the same topic. These examples were not used as training data; they served as in-context signals to help the model understand the expected topic, length, and joke style. This follows the idea of in-context learning, where language models can perform tasks using a small number of examples in the prompt without updating model weights (Brown et al., 2020).

The same prompt structure was used for all topics. The model was asked to generate one short, original, one-line joke clearly related to the topic, appropriate for a general audience, and not copied from the examples. The prompt also instructed the model to avoid offensive, hateful, sexual, or violent content. The temperature was set to 0.8 to balance creativity and coherence. We generated 10 jokes for each topic, resulting in 50 generated jokes in total. This size was chosen to balance topic coverage and the workload of manual evaluation by five raters.

The prompt template was:

You are a humor generation assistant.

Task: Generate ONE short, original, one-line joke clearly related to the topic: [TOPIC].

Audience: General audience.

Style constraints: One-line joke only; natural and clear language; appropriate for a general audience; avoid offensive, hateful, sexual, or violent content; keep it humorous and topic-relevant; avoid repeating common joke patterns; do not copy the examples exactly; use the examples only as style guidance.

Few-shot examples for the topic “[TOPIC]”: Example 1: [EXAMPLE 1]. Example 2: [EXAMPLE 2].

Return only the new joke.

3.6 Evaluation Setup

The evaluation was conducted in three connected stages. First, JokeDetectBERT was used before generation to verify the quality of the few-shot examples. Second, the main evaluation of the generated jokes was conducted through human evaluation. Third, automatic humor evaluation was used as supporting analysis by comparing automatic humor scores with human funniness ratings.

Few-shot example check. Before using the few-shot examples inside the prompt, these examples were checked using VitalContribution/JokeDetectBERT (Hugging Face, 2024). This

model is based on DistilBERT and classifies short text as Joke or No Joke. It was suitable for this step because the examples were short jokes. If the model returned LABEL_1, the example was treated as a joke; if it returned LABEL_0, it was treated as a non-joke. The humor probability was also converted to a 1–5 scale using the formula $S_{\text{humor}} = 1 + (p_{\text{humor}} \times 4)$, where p_{humor} represents the humor probability predicted by the classifier.

Human evaluation. The main evaluation was conducted by five human raters with sufficient English proficiency. Each rater evaluated all generated jokes independently using the same rubric. Each joke was shown with its topic and rated on three criteria: funniness, topic relevance, and fluency. Each criterion used a 1–5 scale. The final score for each joke was calculated as the average of the five raters. This produced Funniness_Avg, Relevance_Avg, and Fluency_Avg.

Supporting automatic evaluation. After human evaluation, two automatic humor classifiers were used as supporting analysis: JokeDetectBERT and mohameddhiab/humor-no-humor (Hugging Face, 2024; Dhiab, 2023). The generated jokes were passed to each model, and the humor probability was converted into a 1–5 score. These automatic scores were compared with human funniness averages using correlation, mean absolute error (MAE), and root mean squared error (RMSE). The goal of this step was not to replace human evaluation, but to examine whether automatic humor classifiers align with human judgment.

4 Results

This section presents the results from three perspectives. First, we report the validation results of the few-shot examples used in the prompting stage. Second, we present the human evaluation results of the generated jokes across the five selected topics. Finally, we compare human funniness judgments with automatic humor scores to examine whether automatic humor classifiers align with human evaluation.

4.1 Few-Shot Example Validation

Before generating new jokes, the selected few-shot examples were evaluated using JokeDetectBERT to check whether the examples used in the prompts had a humorous structure. This step was important

Metric	Result
Total examples	100
Classified as Joke	99
Classified as No Joke	1
No Joke topic	Politics
No Joke sample	Politics_6, Example_2
Overall humor probability	0.9795
Overall humor score	4.918 / 5

Table 3: Few-shot example validation using JokeDetect-BERT. The results show that almost all examples used in the prompts were classified as jokes.

Topic	Funniness	Relevance	Fluency
Animals	3.34	4.70	4.54
Food	3.28	4.68	4.44
Health	3.68	4.96	4.74
Politics	3.90	4.98	4.64
Relationships	3.44	4.78	4.72

Table 4: Human evaluation results by topic. Scores are averaged across five raters using a 1–5 scale.

because the few-shot examples guide the model during generation. If the examples are weak or not humorous, they may negatively affect the quality of the generated jokes.

Out of 100 few-shot examples, 99 were classified as Joke, while only one example was classified as No Joke. The non-joke example belonged to the Politics topic, specifically Sample_ID = Politics_6 and Example_Column = Example_2. It received a humor probability of 0.496 and a humor score of 2.984/5, which is close to the decision boundary.

Overall, the few-shot examples achieved an average humor probability of 0.9795 and an average humor score of 4.918/5. These results indicate that the examples used in the prompts were generally appropriate for guiding GPT-4o-mini during topic-conditioned joke generation.

4.2 Human Evaluation Results

The generated jokes were evaluated by five human raters using three criteria: funniness, topic relevance, and fluency. Each criterion was scored on a 1–5 scale, and the final score for each joke was calculated as the average of the five raters. Table 4 reports the average scores for each topic.

The results show that GPT-4o-mini performed strongest in topic relevance and fluency. Relevance scores were consistently high across all topics,

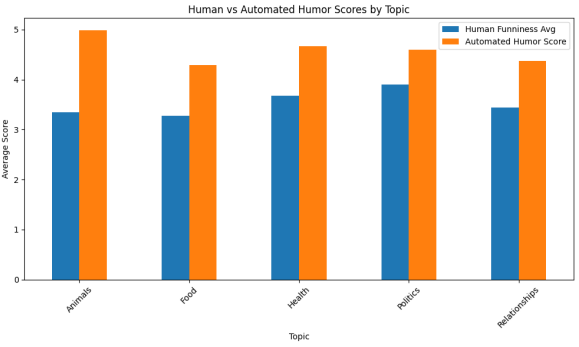


Figure 1: Average human evaluation scores by topic. Relevance and fluency scores remain consistently high across all topics, while funniness scores show greater variation. Politics achieved the highest funniness score, whereas Food achieved the lowest.

ranging from 4.68 to 4.98. This indicates that the model was generally able to follow the requested topic during generation. Fluency scores were also high, ranging from 4.44 to 4.74, showing that the generated jokes were mostly clear, readable, and linguistically natural.

In contrast, funniness scores were lower and more variable than relevance and fluency scores. Politics achieved the highest average funniness score at 3.90/5, followed by Health at 3.68/5. Relationships achieved a moderate funniness score of 3.44/5, while Animals and Food scored lower, with 3.34/5 and 3.28/5 respectively. These results suggest that the model was better at producing topic-relevant and fluent jokes than consistently producing highly funny jokes.

Figure 1 visually confirms the same pattern shown in Table 4. Across all five topics, relevance and fluency remain close to the upper end of the scale, while funniness is noticeably lower. This supports the interpretation that topic control and linguistic quality were easier for the model than generating humor that human raters found strongly funny.

Table 5 summarizes how each research question was answered through the evaluation design. The results show that RQ1 is partially supported: the model produced fluent and understandable jokes, but funniness remained moderate. RQ2 is supported because funniness varied across topics, with Politics and Health scoring higher than Food and Animals. RQ3 is strongly supported because all topics achieved high relevance scores.

Research Question	Purpose	Criteria	Answer Method
RQ1: Can the model generate fluent and humorous jokes?	Measures overall humor generation ability.	Funniness and fluency.	Average funniness and fluency across all generated jokes.
RQ2: Does humor quality vary across topics?	Measures whether topic affects humor quality.	Mainly funniness.	Compare average scores by topic.
RQ3: Can the model follow the requested topic?	Measures topic-following ability.	Topic relevance.	Calculate relevance scores overall and by topic.

Table 5: Research questions and their evaluation criteria.

Model	Corr.	MAE	RMSE
JokeDetectBERT	0.0546	1.2135	1.3676
humor-no-humor	-0.0269	1.7138	1.8740

Table 6: Comparison between automatic humor scores and human funniness scores. Lower MAE and RMSE indicate closer agreement with human ratings.

4.3 Human and Automatic Evaluation Comparison

To examine whether automatic humor models align with human judgments, we compared automatic humor scores with the average human funniness scores. Two automatic humor classifiers were used: JokeDetectBERT and mohameddhiab/humor-no-humor. The comparison was based on correlation, mean absolute error (MAE), and root mean squared error (RMSE). Table 6 summarizes the results.

The results show that JokeDetectBERT was closer to human ratings than mohameddhiab/humor-no-humor, as indicated by its lower MAE and RMSE. However, both models showed very weak correlation with human funniness scores. JokeDetectBERT achieved a correlation of only 0.0546, while mohameddhiab/humor-no-humor achieved a slightly negative correlation of -0.0269.

This suggests that automatic humor classifiers may detect whether a generated text resembles a joke, but they do not reliably measure how funny humans actually find it. Therefore, automatic evaluation can be useful as a supporting analysis, but it cannot replace human evaluation in humor generation tasks.

4.4 Summary of Findings

Overall, the results show three main findings. First, the few-shot examples were mostly valid, with 99 out of 100 examples classified as jokes by JokeDe-

tectBERT. This supports the quality of the examples used to guide the generation process. Second, GPT-4o-mini generated jokes with strong topic relevance and fluency, but only moderate funniness. This means that the model was effective at following topic conditions and producing readable language, but humor quality remained harder to achieve. Third, automatic humor classifiers showed weak agreement with human funniness judgments. Although JokeDetectBERT performed better than mohameddhiab/humor-no-humor, both models failed to strongly align with human perception of humor.

5 Discussion

The results show that topic-guided humor generation using GPT-4o-mini can produce fluent and topic-relevant jokes, especially when the generation process is controlled through few-shot prompting. The high relevance and fluency scores indicate that the model was generally able to follow the requested topics and generate linguistically understandable outputs. However, funniness remained more variable across topics, which is expected because humor depends on personal taste, cultural background, social context, and the strength of the punchline.

One important finding is that relevance and fluency were consistently higher than funniness. This means that generating a grammatically correct and topic-relevant sentence is not enough to guarantee successful humor. This finding is consistent with previous work showing that humor generation remains difficult even for advanced neural and large language models (Amin and Burghardt, 2020; Jentsch and Kersting, 2023). In this project, GPT-4o-mini often understood the topic and produced readable jokes, but human raters did not always judge the outputs as highly funny. This supports the view that humor generation requires more than

surface-level fluency or topic alignment.

The results also highlight the importance of human evaluation in humor-related NLP tasks. The weak correlation between automatic humor scores and human funniness ratings suggests that automatic humor classifiers may detect joke-like patterns, but they do not reliably measure actual perceived funniness. This agrees with broader work on natural language generation evaluation, which argues that automatic metrics often fail to capture subjective qualities such as creativity, coherence, and user perception (Celikyilmaz et al., 2020; van der Lee et al., 2021). Therefore, in this project, automatic humor models were useful as supporting tools, but human evaluation remained the main basis for judging joke quality.

From a social perspective, this work shows that humor generation should not be treated as a simple entertainment task. Humor can be useful in education, marketing, entertainment, and conversational systems, but it can also become offensive, harmful, or socially inappropriate if the data and prompts are not controlled. The results of this project suggest that safer humor generation requires multiple layers of control, including data cleaning, toxicity filtering, topic selection, prompt safety instructions, and human-centered evaluation. These steps can help reduce the risk of generating offensive jokes and make humor generation more suitable for real-world user-facing applications.

The social value of this project is also connected to responsible AI design. As generative language models become more common in chatbots, learning platforms, and digital assistants, they may increasingly be used to produce casual, humorous, or persuasive text. This makes it important to understand not only whether a model can generate jokes, but whether it can generate jokes that are relevant, readable, and appropriate for the intended audience. By evaluating humor across different topics and comparing human judgments with automatic scores, this work contributes to the development of more careful and socially aware generation systems.

However, this project has several limitations. First, the human evaluation involved only five raters, which is not enough to represent broad differences in humor taste, culture, language background, or age. Second, only five topics were selected, so the results cannot be generalized to all humor categories or social contexts. Third, the generated set contained only 50 jokes, which is suitable for a course project but limited for broader statis-

tical conclusions. Fourth, the automatic humor detection models were trained on specific datasets and may not capture all types of humor or cultural variation. Fifth, although ethical filtering and safety instructions were used, the model may still generate unsuitable content in other settings. Finally, the project did not compute inter-rater agreement, which would provide a stronger measure of rating consistency.

Despite these limitations, the study provides a structured and manageable NLP experiment. It shows that topic-guided humor generation can be studied through a controlled pipeline that combines topic discovery, topic mapping, few-shot prompting, human evaluation, and supporting automatic analysis. The findings suggest that GPT-4o-mini is strong at producing relevant and fluent jokes, but that funniness remains harder to control and evaluate.

6 Conclusion

This project presented an applied study on topic-guided joke generation using GPT-4o-mini. The work started from the Short Jokes Dataset, then the data was cleaned, reduced, and filtered to remove inappropriate or low-quality texts. Topic discovery was used to identify common themes in the dataset, followed by topic mapping to merge similar labels and select five final content-based categories for generation: Relationships, Health, Food, Animals, and Politics.

The generation process relied on few-shot prompting, where the model was provided with two humorous examples related to each topic before generating new jokes. Before using these examples, they were checked automatically using JokeDetect-BERT to ensure that they had a humorous structure. The generated jokes were then evaluated by five human raters using three criteria: funniness, topic relevance, and fluency. Automatic humor detection models were also used as supporting tools to compare their scores with human funniness judgments.

The results showed that GPT-4o-mini performed well in topic relevance and fluency, indicating that it could generate understandable jokes related to the selected topics. However, funniness was more variable and generally lower than relevance and fluency, confirming that humor remains a difficult and subjective generation task. The comparison with automatic humor classifiers also showed weak alignment with human judgment, meaning that au-

tomatic evaluation can support analysis but cannot replace human evaluation.

Overall, this project shows that topic-guided humor generation is a promising direction in NLP when it is handled through a responsible, controlled, and human-centered evaluation process. The study also highlights the importance of safety filtering and topic control when generating humorous content for user-facing applications. Future work could expand the experiment by using more topics, more raters, larger generated samples, different language models, and additional evaluation criteria such as originality, surprise, cultural appropriateness, and offensiveness.

Future studies could also compute inter-rater reliability and compare zero-shot with few-shot prompting to better understand how prompting strategies affect humor quality.

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